TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	91.0 ± 0.89
Correlation Stability	92.8 ± 0.40
Distribution Consistency	92.2 ± 1.66
Entropy Preservation	80.6 ± 5.50
Mean Z Score Drift	0.1 ± 0.01
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	17.5 ± 1.91
Data Density	100.0 ± 0.00
Coverage Score	67.6 ± 3.56
Feature Accuracy	75.6 ± 7.94
Novelty Score	0.7 ± 0.02

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Dataset Audit: thoracic_surgery.csv

1. Dataset Portrait

Attribute	Value
Rows	470
Columns	17
Numeric features	17
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
DGN	3.096	3.109	+0.013	0.4%	OK
PRE4	3.282	3.252	-0.029	0.9%	OK
PRE5	4.569	4.465	-0.104	2.3%	OK
PRE6	0.781	0.838	+0.057	7.3%	OK
PRE7	0.066	0.074	+0.009	12.9%	Warn
PRE8	0.145	0.168	+0.023	15.8%	Warn
PRE9	0.066	0.061	-0.005	7.3%	OK
PRE10	0.687	0.737	+0.049	7.2%	OK
PRE11	0.166	0.165	-0.001	0.6%	OK
PRE14	11.736	11.747	+0.011	0.1%	OK
PRE17	0.074	0.096	+0.021	28.6%	Drift
PRE19	0.004	0.013	+0.009	212.5%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	91.0 ± 0.89
Correlation Stability (*2)	92.8 ± 0.40
Distribution Consistency (*3)	92.2 ± 1.66
Entropy Preservation (*4)	80.6 ± 5.50
Mean Z Score Drift (*5)	0.1 ± 0.01
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	17.5 ± 1.91

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	67.6 ± 3.56
Feature Accuracy (*10)	75.6 ± 7.94
Novelty Score (*11)	0.7 ± 0.02

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8542553191489363	0.8510638297872343	0.8563829787234043	0.00212765957446811 03	
F1	N/A	0.011764705882352941	0.026050420168067225	0.02605042016806722 5	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8138297872340425	0.8127659574468085	0.8127659574468085	-0.00106382978723403 85	
F1	0.11972345964178996	0.08625206177837756	0.06764103440419231	-0.05208242523759765	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.21063829787234045	0.2393617021276596	0.2936170212765957	0.08297872340425533	
F1	0.256855620661189	0.21224090582103577	0.23378975522000999	-0.02306586544117913	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8117021276595745	0.7946808510638298	0.7978723404255319	-0.01382978723404257 8	
F1	0.12873355257087923	0.12005840028265657	0.10623074058557928	-0.02250281198529996 4	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8457446808510639	0.8521276595744682	0.8542553191489363	0.00851063829787236 3	
F1	0.011764705882352941	0.0	N/A	-0.01176470588235294 1	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.847872340425532	0.8531914893617023	0.8542553191489363	0.00638297872340425	
F1	N/A	0.009523809523809523	N/A	0	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8382978723404255	0.8297872340425533	0.8436170212765959	0.005319148936170215	
F1	0.08705128205128206	0.04166666666666664	0.06088336783988958	-0.026167914211392473	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8553191489361703	0.8553191489361703	0.8531914893617023	-0.002127659574468095	
F1	N/A	0.0	N/A	0	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8553191489361703	0.8553191489361703	0.8553191489361703	0	
F1	N/A	0.0	N/A	0	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.851063829787234	0.85	0.8553191489361703	0.004255319148936154	
F1	0.037675637056442016	0.0	N/A	-0.03767563705644201	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	2.0	2.88	2.16	1.0	0.0	0.0	0.0	1.0	1.0	14.0
2	3.0	3.4	1.88	0.0	0.0	0.0	0.0	0.0	0.0	12.0
3	3.0	2.76	2.08	1.0	0.0	0.0	0.0	1.0	0.0	11.0
4	3.0	3.68	3.04	0.0	0.0	0.0	0.0	0.0	0.0	11.0
5	3.0	2.44	0.96	2.0	0.0	1.0	0.0	1.0	1.0	11.0
6	3.0	2.48	1.88	1.0	0.0	0.0	0.0	1.0	0.0	11.0
7	3.0	4.36	3.28	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	2.0	3.19	2.5	1.0	0.0	0.0	0.0	1.0	0.0	11.0
9	3.0	3.16	2.64	2.0	0.0	0.0	0.0	1.0	1.0	11.0
10	3.0	2.32	2.16	1.0	0.0	0.0	0.0	1.0	0.0	11.0
11	3.0	2.56	2.32	0.0	0.0	1.0	0.0	1.0	0.0	12.0
12	3.0	4.28	4.44	1.0	0.0	0.0	0.0	0.0	0.0	12.0
13	3.0	3.0	2.36	1.0	0.0	0.0	0.0	1.0	1.0	11.0
14	2.0	3.98	3.06	2.0	0.0	0.0	0.0	1.0	1.0	14.0
15	3.0	1.96	1.4	1.0	0.0	0.0	0.0	1.0	0.0	11.0
16	3.0	4.68	4.16	1.0	0.0	0.0	0.0	1.0	0.0	12.0
17	2.0	2.21	1.88	0.0	0.0	1.0	0.0	0.0	0.0	12.0
18	2.0	2.96	1.67	0.0	0.0	0.0	0.0	0.0	0.0	12.0
19	3.0	2.6	1.68	1.0	0.0	0.0	0.0	1.0	0.0	12.0
20	3.0	2.88	2.48	0.0	0.0	0.0	0.0	0.0	0.0	11.0

B.1 Audit Trace: Fold 1 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.36	1.76	1.0	0.0	1.0	1.0	1.0	1.0	11.0
2	3.0	3.64	2.72	1.0	0.0	0.0	0.0	0.0	0.0	12.0
3	3.0	2.48	1.88	0.0	0.0	1.0	0.0	0.0	0.0	12.0
4	3.0	2.5	2.04	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	2.0	2.44	2.24	1.0	0.0	0.0	0.0	1.0	0.0	11.0
6	3.0	4.32	3.6	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	3.0	2.68	2.16	0.0	0.0	0.0	0.0	0.0	0.0	12.0
8	4.0	2.32	1.76	1.0	0.0	1.0	0.0	1.0	1.0	11.0
9	2.0	2.72	2.08	0.0	0.0	0.0	0.0	0.0	0.0	12.0
10	3.0	2.32	1.76	0.0	0.0	0.0	0.0	0.0	0.0	12.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	4.32	3.36	1.0	0.0	0.0	0.0	1.0	0.0	11.0
2	3.0	2.36	1.56	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	2.0	4.6	3.2	1.0	0.0	0.0	0.0	1.0	0.0	11.0
4	2.0	2.64	2.2	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	3.0	2.96	2.24	1.0	0.0	0.0	0.0	1.0	0.0	11.0
6	3.0	3.64	2.72	1.0	0.0	0.0	0.0	1.0	0.0	12.0
7	3.0	2.84	2.24	1.0	1.0	1.0	0.0	0.0	0.0	12.0
8	3.0	3.36	2.8	1.0	0.0	0.0	0.0	1.0	1.0	12.0
9	3.0	2.44	1.96	1.0	0.0	1.0	1.0	1.0	0.0	13.0
10	3.0	5.16	4.28	0.0	0.0	0.0	0.0	0.0	0.0	12.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	8.0	5.08	86.3	1.0	1.0	0.0	0.0	1.0	1.0	12.0
2	3.0	2.8	2.52	1.0	0.0	0.0	0.0	1.0	0.0	14.0
3	3.0	2.32	2.08	1.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	2.16	1.44	2.0	0.0	0.0	0.0	1.0	0.0	12.0
5	2.0	2.56	2.36	1.0	0.0	0.0	0.0	1.0	0.0	11.0
6	3.0	2.84	2.32	1.0	0.0	0.0	0.0	0.0	1.0	12.0
7	4.0	4.24	3.68	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	3.0	3.87	2.68	0.0	0.0	0.0	0.0	0.0	0.0	12.0
9	5.0	3.67	76.8	0.0	1.0	1.0	0.0	1.0	0.0	12.0
10	3.0	3.8	2.84	1.0	0.0	0.0	0.0	1.0	0.0	12.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	2.0	3.8	3.16	1.0	0.0	0.0	0.0	1.0	1.0	11.0
2	3.0	3.6	2.64	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	2.0	2.7	2.15	1.0	0.0	1.0	0.0	1.0	0.0	11.0
4	5.0	3.96	3.56	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	2.0	2.94	1.72	1.0	0.0	0.0	0.0	0.0	0.0	11.0
6	5.0	5.2	3.96	0.0	0.0	0.0	0.0	0.0	0.0	12.0
7	3.0	1.84	1.28	1.0	0.0	0.0	0.0	1.0	1.0	11.0
8	5.0	4.95	4.12	1.0	0.0	0.0	0.0	0.0	1.0	11.0
9	3.0	2.92	2.28	1.0	0.0	0.0	0.0	1.0	0.0	11.0
10	3.0	2.32	1.76	0.0	0.0	0.0	0.0	0.0	0.0	12.0

B.5 Audit Trace: Fold 5 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	5.0	2.0	1.6	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	4.0	5.36	67.3	0.0	0.0	0.0	0.0	0.0	0.0	11.0
3	3.0	4.08	3.2	0.0	0.0	0.0	0.0	0.0	0.0	12.0
4	3.0	3.04	2.21	2.0	0.0	0.0	0.0	0.0	1.0	11.0
5	2.0	3.44	2.88	1.0	0.0	0.0	0.0	0.0	0.0	11.0
6	3.0	3.72	2.96	1.0	0.0	0.0	0.0	1.0	1.0	12.0
7	3.0	3.0	2.44	1.0	0.0	0.0	0.0	1.0	1.0	12.0
8	3.0	2.4	1.88	1.0	0.0	0.0	0.0	1.0	0.0	11.0
9	3.0	4.36	3.92	1.0	0.0	0.0	0.0	0.0	0.0	11.0
10	3.0	3.4	2.8	1.0	1.0	0.0	0.0	1.0	1.0	11.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	1.84	1.32	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	4.56	2.96	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	3.0	2.72	1.72	1.0	0.0	0.0	0.0	1.0	0.0	11.0
4	3.0	2.73	2.28	0.0	0.0	0.0	0.0	0.0	0.0	12.0
5	3.0	3.76	1.64	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	3.0	2.8	1.76	1.0	0.0	0.0	0.0	1.0	0.0	11.0
7	2.0	3.24	2.52	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	3.0	2.4	1.64	0.0	0.0	0.0	0.0	0.0	0.0	11.0
9	3.0	4.28	4.44	1.0	0.0	0.0	0.0	0.0	0.0	12.0
10	3.0	3.24	52.3	0.0	0.0	0.0	0.0	0.0	0.0	12.0

B.7 Audit Trace: Fold 7 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.6	1.96	0.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	2.6	2.32	0.0	0.0	0.0	0.0	0.0	0.0	12.0
3	3.0	4.68	3.44	0.0	0.0	0.0	0.0	1.0	0.0	12.0
4	2.0	4.0	2.94	1.0	0.0	0.0	0.0	1.0	0.0	12.0
5	2.0	3.44	2.76	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	3.0	3.24	2.92	1.0	0.0	1.0	0.0	1.0	0.0	11.0
7	3.0	2.84	2.16	1.0	0.0	0.0	0.0	1.0	0.0	12.0
8	3.0	2.88	2.48	0.0	0.0	0.0	0.0	0.0	0.0	11.0
9	3.0	2.2	1.44	1.0	0.0	0.0	0.0	1.0	0.0	12.0
10	4.0	5.16	4.96	1.0	0.0	0.0	0.0	0.0	0.0	11.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	2.96	2.32	1.0	0.0	0.0	0.0	1.0	0.0	12.0
2	3.0	2.96	2.0	1.0	0.0	0.0	0.0	1.0	0.0	12.0
3	3.0	2.44	1.88	1.0	0.0	0.0	0.0	0.0	0.0	12.0
4	3.0	2.92	2.2	1.0	0.0	0.0	0.0	1.0	1.0	11.0
5	5.0	2.88	2.16	1.0	0.0	0.0	0.0	0.0	0.0	12.0
6	4.0	2.52	3.92	0.0	1.0	0.0	0.0	0.0	0.0	12.0
7	3.0	2.76	2.28	0.0	0.0	0.0	0.0	0.0	0.0	11.0
8	4.0	5.03	79.3	1.0	0.0	0.0	1.0	0.0	0.0	11.0
9	2.0	2.16	1.56	1.0	0.0	0.0	0.0	1.0	0.0	11.0
10	3.0	2.0	1.0	1.0	0.0	1.0	0.0	1.0	1.0	11.0

B.9 Audit Trace: Fold 9 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	2.0	3.68	2.36	1.0	0.0	0.0	0.0	1.0	1.0	12.0
2	3.0	2.24	1.0	1.0	0.0	0.0	0.0	1.0	0.0	11.0
3	3.0	2.92	2.08	0.0	0.0	0.0	0.0	0.0	0.0	12.0
4	3.0	3.52	52.3	0.0	0.0	1.0	0.0	0.0	0.0	11.0
5	4.0	2.72	1.96	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	3.0	5.2	3.28	0.0	0.0	0.0	1.0	1.0	0.0	11.0
7	3.0	3.44	3.13	1.0	0.0	0.0	0.0	1.0	1.0	12.0
8	3.0	3.64	3.12	1.0	0.0	0.0	0.0	1.0	0.0	12.0
9	3.0	2.56	60.9	0.0	0.0	0.0	0.0	0.0	0.0	11.0
10	3.0	2.8	3.36	1.0	0.0	0.0	0.0	1.0	0.0	12.0

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B.10 Audit Trace: Fold 10 | Gen: 100%

#	DGN	PRE4	PRE5	PRE6	PRE7	PRE8	PRE9	PRE10	PRE11	PRE14
1	3.0	4.2	3.78	1.0	0.0	0.0	1.0	1.0	0.0	12.0
2	3.0	1.96	1.44	2.0	0.0	0.0	0.0	1.0	1.0	11.0
3	3.0	2.64	2.2	0.0	0.0	0.0	0.0	1.0	0.0	12.0
4	3.0	3.48	2.82	0.0	0.0	0.0	0.0	0.0	0.0	12.0
5	3.0	2.0	1.6	1.0	0.0	0.0	0.0	1.0	0.0	12.0
6	3.0	3.52	2.44	1.0	0.0	0.0	0.0	1.0	0.0	11.0
7	3.0	3.48	2.52	1.0	0.0	0.0	0.0	1.0	0.0	14.0
8	3.0	4.48	3.48	0.0	0.0	0.0	0.0	0.0	0.0	12.0
9	3.0	2.63	67.3	1.0	0.0	0.0	1.0	1.0	0.0	11.0
10	4.0	5.16	4.96	1.0	0.0	0.0	0.0	0.0	0.0	11.0